Band-1 Hadamard Order 4612 — Miyamoto (Paley-I seed H_1152); Gap 2052 Still OPEN

H_1152/C_1154/H_4612 PASS; 4612 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4516_Miyamoto`; `2026-09-15_Band1_Hadamard_4372_Miyamoto` (Paley-I seed pattern).
Lab. `/workspace/experiments/band1/hadamard_4612_miyamoto/`.
Cutoff. 2026-09-15T16:51:15Z (~12:51PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1152 and 4612 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I seed H_{1152} (H_{1152} = Paley-I($q = 1151$)). Paley-II path for H_{1152} is **SKIP** ($q = 575 = 5^2 \cdot 23$ not prime). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4612 is a **Miyamoto-only bonus (NOT in SPK)**: Paley-I **FAIL** ($q = 4611 = 3 \cdot 29 \cdot 53$ not prime); Paley-II **FAIL** ($q = 2305 = 5 \cdot 461$ not prime); Kronecker **FAIL** ($4612 = 4 \cdot 1153$, 1153 not Hadamard-admissible). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4516 on box. This increment ships **order 4612** ($4612 = 4 \cdot 1153$, 1153 prime $\equiv 1 \pmod 4$) via Miyamoto with a **Paley-I seed** H_{1152}. Symmetric Paley conference C_{1154} over prime 1153; $H_{4612} = \text{Miyamoto}(H_{1152}, C_{1154})$. Unit tests **46/46 PASS**. SPK next gap remains **2052** (OPEN). Order 4612 is **not** in SPK (Miyamoto-only bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4612; Miyamoto; Paley type I; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{1152} constructed and verified (Paley type I, $q = 1151$). Paley-II path for H_{1152} **SKIP** ($q = 575 = 5^2 \cdot 23$ not prime). 2. C_{1154} constructed and verified (Paley conference, $q = 1153$): $CC^\top = 1153I$, zero diagonal. 3. H_{4612} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 4. SPK: $4612 \notin$ SPK (Miyamoto-only). Paley-I/II/Kronecker paths for 4612 **rejected**. Next constructive gap remains **2052** (OPEN). Verified H_{1152} enters SPK as classical Paley-I seed.

2. Arithmetic and eligibility

Quantity	Value
Target order	$4612 = 4 \cdot 1153$
$q = 1153$	prime, $1153 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1152}
Needed conference	order $1154 = q + 1$
H_{1152} path (used)	Paley-I: $H_{1152} = \text{Paley-I}(q = 1151)$
H_{1152} Paley-II	**SKIP**: $2(q+1) = 1152 \Rightarrow q = 575 = 5^2 \cdot 23$ not prime
Conference path	Symmetric Paley over prime 1153
Paley-I for 4612	**REJECTED**: $q' = 4611 = 3 \cdot 29 \cdot 53$ not prime
Paley-II for 4612	**REJECTED**: $q' = 2305 = 5 \cdot 461$ not prime
Kronecker for 4612	**REJECTED**: only 4×1153; $1153 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{1152}	Paley type I, $q = 1151$	0	1.4994	**PASS**
H_{1152} Paley-II	$q = 575 = 5^2 \cdot 23$	—	—	**SKIP** (not prime)
C_{1154}	Paley conference, $q = 1153$	0	1.3108	**PASS**
H_{4612}	Miyamoto(H_{1152}, C_{1154})	0	47.1582	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 51.5666 s. Compressed ± 1 matrices saved as `H1152_pm1.npz`, `C1154_pm1.npz`, `H4612_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior classical/SPK ships (through 4036/3908/3748/… + **1020/1116** Paley-II + **1032/1092** Paley-I + **524/1048/548/1096/564/1128** Double Paley-I; **exclude** 4516/4468/4388/4372/4196/4132/4084/3716/3524/3284/3076/3044/2836/2692/2564 Miyamoto-only) $\cup$ verified H_{1152} (Paley-I) $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4700). **Miyamoto is not folded into SPK**; order 4612 remains a Miyamoto-only bonus.

Quantity	Value
$4612 \in$ SPK?	**False** (Miyamoto-only bonus)
$1152 \in$ SPK?	**True** (Paley-I classical)
Next constructive gap	**2052** (OPEN)
SPK path audit	Paley-I FAIL; Paley-II FAIL; Kronecker FAIL

5. Gap 2052 (method triage / non-finding ONLY)

Standard constructive gap **2052** remains **OPEN**. Paley-I/II/Kronecker and Miyamoto-with-Paley-conference paths fail on box (conference prime $q = 513$ not prime). GS/SDS / Yamada / Williamson–Turyn remain lit-only this run. Non-finding $\neq$ counterexample. Conjecture **OPEN**.

6. Claim level and non-claims

Claim. Finite constructions of orders **1152** and **4612** ONLY (verified).
Non-claims. Full Hadamard conjecture; Clay-closed; finished-ToE; RH; P$\neq$NP; YM; Frankl; NS-solved; AC-solved; gap 2052 filled.
GATE-04 collection **LOCKED**. `clay_closed=false`.

7. Next optional targets

- $q = 1181 \to n = 4724$ need H_{1180}: (no classical seed note)
- $q = 1193 \to n = 4772$ need H_{1192}: (no classical seed note)
- $q = 1201 \to n = 4804$ need H_{1200}: Double(PaleyI half=600), Kronecker(4x300)
- $q = 1213 \to n = 4852$ need H_{1212}: (no classical seed note)
- $q = 1217 \to n = 4868$ need H_{1216}: Double(PaleyI half=608), Kronecker(4x304)
- $q = 1229 \to n = 4916$ need H_{1228}: PaleyII(q=613)
- $q = 1237 \to n = 4948$ need H_{1236}: PaleyII(q=617)
- $q = 1249 \to n = 4996$ need H_{1248}: Kronecker(4x312)

Also: method-attack SPK gap **2052**, or Track B privacy / Object 1 meter await.

8. Artifacts

- `verify_hadamard_4612.py`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `run.log`
- `H1152_pm1.npz`, `C1154_pm1.npz`, `H4612_pm1.npz`
- Lab note PDF / SCOREBOARD / STATUS_BOARD on `/workspace/DOWNLOAD_PDFs/`

End. overall_pass=True; units=46/46; wall_total=51.5666s; GATE-04 LOCKED; conjecture OPEN; gap 2052 OPEN; 4612 Miyamoto-only bonus.